CO-1 Maths
Home Assignment-1

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1) a) $Sk = \frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$

$= \frac{3(68-61)}{14} = \frac{3\times7}{42} = 1.5$

co-efficient of skewers $= 1.5$

b) Mean > Median

∴ the distribution is +vely (right) skewers.

c) For kurtusis:

⇒ <3 → playkurtic

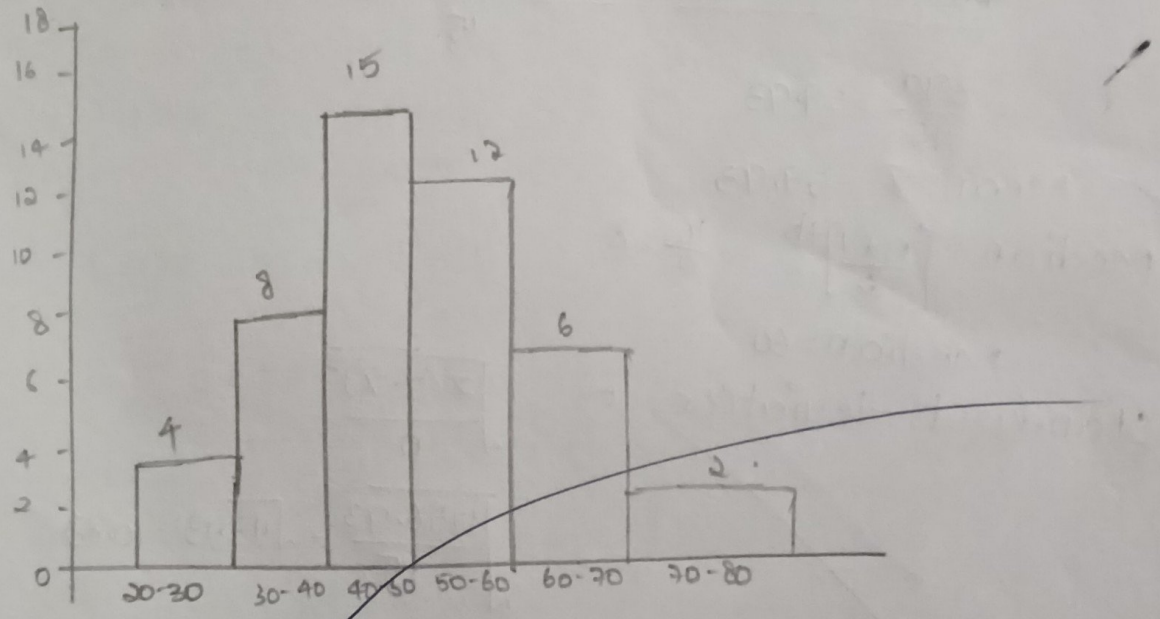
⇒ $=3$ → mesokurtic

⇒ >3 → upto kurtic

Given: kurtosis $= 4.8 > 3$

The distribution is ceptokurtic.

2) Histogram:



b) Mean $\bar{x} = \frac{\Sigma Rimi}{\Sigma Ri}$ $\quad [mi = \frac{u+u_1}{2}]$

$= \frac{(25\times4)+(35\times8)+(45\times15)+(55\times10)+(65\times6)+(15\times4)}{45}$

$= \frac{100+280+675+550+390+150}{45}$

$= \frac{2145}{45} = 47.67$

$\bar{x} = 47.67$

Median $= L + (\frac{N}{2} - cf)\times h$

$= 40 + \left[\frac{225-b}{15}\right] \times 10$

$= 40 + \left[\frac{10.5}{5}\right] \times 10$

$= 40 + \left[\frac{7}{10}\right] \times 10$

$= 40 + 7 = 47$

Median = 47

Mean > Median.

∴ The distribution is appropriately +vely skewed.

c) The mean is likely to be larger than the median. In a +vely skewed distribution, the long right tail pulls the mean towards higher values, making it greater than the median.

3) Given,

42, 45, 48, 50, 51, 55, 58, 60, 62, 65, 67, 70, 72, 75, 78.

mean $\bar{x} = \frac{\Sigma x_i}{n}$

$= \frac{42+45+48+50+51+55+58+60+62+65+67+70+72+75+78}{15}$

$= \frac{899}{15} = 59.93$

Mean $\bar{x} = 59.93$

Median $= \left[\frac{n+1}{2}\right]^{th} = \frac{16}{2} = 8$

Median = 60

Standard deviation $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$

$= \sqrt{\frac{1756.93}{15}} = \sqrt{117.13} = 10.82$

$\boxed{\sigma = 10.82}$

4) Given 120, 125, 128, 130, 132, 135, 138, 140, 145, 150

Mean $\bar{x} = \frac{\Sigma x_i}{n}$

$= \frac{120+125+128+130+132+135+138+140+145+150}{10}$

$= \frac{1343}{10} = 134.3$

$\bar{x} = 134.3$

Variance $\sigma^2 = \frac{\Sigma(x-\bar{x})^2}{n}$

$$= \frac{204.49 + 86.49 + 39.69 + 18.49 + 5.29 + 0.49 + 13.69 + 32.49 + 114.49 + 246.49}{10}$$

$$= \frac{762.1}{10} = 76.21$$

$$\sigma^2 = 76.21$$

Standard Deviation $\sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}}$

$$= \sqrt{76.21}$$

$$\boxed{\sigma = 8.72}$$

5) Given,

minimum = 15

$Q_1 = 25$

Median = 35

$Q_3 = 48$

maximum = 75

$IQR = Q_3 - Q_1 = 18 - 25 = 23$

$$\boxed{IQR = 23}$$

Lower Hence $= Q_1 - 1.5\,IQR$

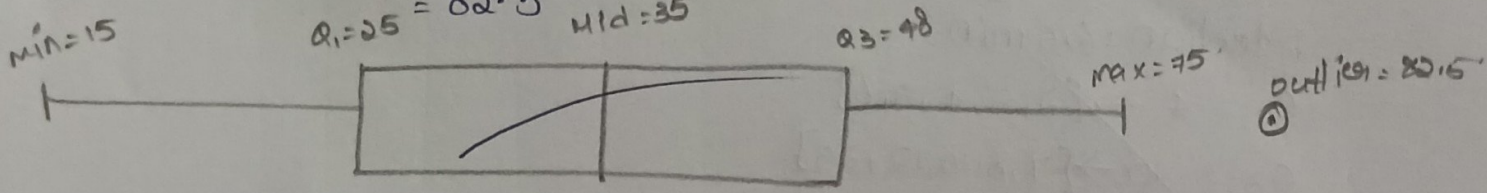
$= 25 - 34.5$

$= -9.5$

Upper Fence $= Q_1 + 1.5\,IQR$

$= 48 + 34.5$

$= 82.5$



6) Mean $= \frac{\Sigma x_i}{n} = \frac{3.5 + 4.0 + 4.2 + 4.5 + 4.8 + 5.0 + 5.3 + 5.5 + 18.0 + 22.0}{10}$

$= \frac{72.8}{10} = 7.28$ lakh

Median $= \frac{\left[\frac{n}{2}\right]^{th} + \left[\frac{n+1}{2}\right]^{th}}{2} = \frac{4.8 + 5.0}{2} = \frac{9.8}{2}$

Median = 4.9 lakh

The median better represents this default the data contains two extreme values (18.08 22.0) which skew the mean upwards it high for majority of employees.

7) Mean = $\frac{\Sigma x_i}{n} = \frac{45+48+58+52+55+58+60+62+65+120}{10}$

$= \frac{615}{10} = 61.5$

Mean = 61.5

Median = $\frac{\left[\frac{n}{2}\right]^{th} + \left[\frac{n+1}{2}\right]^{th}}{2} = \frac{55+58}{2} = \frac{113}{2}$

Median = 56.5

standard deviation (σ) = $\sqrt{\frac{\Sigma (x-\bar{x})^2}{n}}$

$= \frac{272.25+182.25+132+25+90.25+42.25+12.25+2.25+0.25+12.25+3422.25}{10}$

$= \frac{4102.5}{10} = 410.25$ SD $\sigma = \sqrt{410.25} = 20.25$

standard deviation = 20.25

The value 120 is an extreme value compared to rest of dataset it distorts mean, making median a better measure of central tendency.

8) Mean = $\frac{\Sigma x_i}{n} = \frac{2+3+4+5+6+7+8+10+12+25}{10}$

$= \frac{82}{10} = 8.2$ min

Mean = 8.2 min

Median = $\frac{\left[\frac{n}{2}\right]^{th} \left[\frac{n+1}{2}\right]^{th}}{2} = \frac{6+7}{2} = 6.5$ min

Median = 6.5 min

$Q_1 = 4 \rightarrow [2, 3, 4, 5, 6]$

$Q_3 = 10 \rightarrow [7, 8, 10, 12, 25]$

IQR = $Q_3 - Q_1 = 10 - 4 = 6$.

Mean > Median

The distribution 4 positively skewed since 25 > 19, 25mm is a outlier using 1.5 · IQR · U.B = $Q_3 + (1.5 \times IQR) = 10 + (1.5 \times 6) = 19$.

9) Histogram & box plots show set use different visual summaries.

Histograms: Use adjacent vertical bars to represent data frequency within specific numerical intervals (bins)

Box plot: use a central rectangle (box) and extended lines (whiskers) to map data distribution.

b) Basic steps involved in exploratory Data Analysis (EDA)

- Understand the dataset.
- Inspect structures and data types
- check missing values
- Generate summary stats.
- Study Distribution.
- Reflect outliers.
- Analyse relationship.
- visualize the data.
- Draw initial insights.

7.7.26